# FusePlan: Separating Complementarity from Ensemble Effects in Partial Floor Plan Retrieval

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## Abstract

*Architects often search for floor plans starting from an incomplete layout, yet floor-plan retrieval methods train and evaluate on complete plans. We study retrieval from incomplete queries on RPLAN with FusePlan, which fuses a graph branch and a vision branch over the same gallery of 67,405 plans. The graph branch, a GATv2 encoder trained to map plans with rooms removed onto complete plans, recovers the source plan with an AUC of 0.903 on the test split (the reciprocal rank of the source averaged over three fractions of removed rooms; 1 means always first), against 0.611 for LayoutGKN retrained on the same task. A fusion gain alone does not show that the image adds information, because fusing any two models helps. We therefore compare each fusion with ensemble controls that fuse the graph with a second graph model, and define the complementarity gain as the part of the gain the controls do not reach. With this strong graph, a frozen image encoder adds* $+0.013$ *AUC and no complementarity gain, while vision heads trained on the damage pass both controls by* $+0.012$ *to* $+0.022$ *over four replicas. The fusion gain peaks when the two branches are equally strong, and the image helps most where the graph keeps few rooms. We make every choice on the validation split and read the test split with pre-registered predictions.*

## 1. Introduction

Architects rarely start from a complete floor plan. They sketch a few rooms, fix an outline or reuse a fragment of an existing plan, and then look for complete plans that contain it. A retrieval system for this workflow must rank a large collection of plans from an *incomplete* query and still find the plan the fragment came from.

Two gaps separate current methods from this workflow. First, floor-plan similarity methods learn and evaluate on complete plans: LayoutGMN [13] and LayoutGKN [20] learn a structural similarity between complete RPLAN plans and score it with triplet accuracy or precision among their own top results. Partial plans appear in floor-plan generation [9, 23], and we found no retrieval method that takes them as queries. Second, a plan exists both as an image and as a graph of rooms, and fusing the two looks attractive because they fail in different ways: the image keeps the shape of the hole a removed room leaves, while a partial graph is the plausible complete graph of many smaller plans. Yet any fusion of two models with partly independent errors gains accuracy; a fusion gain alone cannot tell whether the image adds information the graph lacks or merely acts as a second model.

We separate the two with *ensemble controls*: we fuse the graph with a second model that holds the same information, either a second training of the same graph encoder or an-

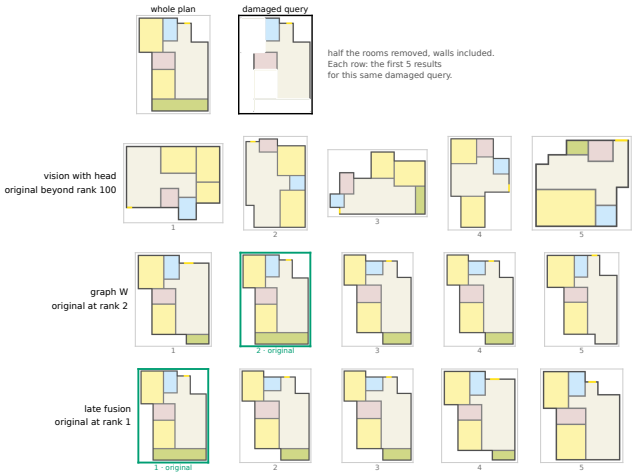



Figure 1. **Retrieving a plan from half of its rooms.** Top 5 of each system for one validation query; a frame marks the source. The single-damage vision head misses it, the graph ranks it second behind a near-identical plan, and the fusion ranks it first. The query is the first, alphabetically, of the 34 validation queries at this level where only the fusion ranks the source first.

other graph encoder. We call *complementarity gain* (CG) the part of the vision–graph gain that the stronger control does not reach. A positive CG means the image contributes information of its own.

FusePlan has two retrieval branches over the same gallery of 67,405 RPLAN plans [22] (Fig. 2). The graph branch is a GATv2 encoder [5] trained to map a plan with rooms removed onto the complete plan, with the checkpoint chosen on retrieval robustness. The vision branch is a frozen PE-Spatial encoder [4] with whitening, optionally followed by a head trained on the same damages. A late fusion with a single weight combines their cosine similarities, and the two branches and the fusion rank the gallery with the same exact inner-product search. We measure robustness as the self-recovery of the source plan when we remove 25, 50 or 75% of its rooms, or of its area as a region or as image patches: a query scores the reciprocal rank of its source, and the AUC averages it over the three levels, so 1 means the source always ranks first (Sec. 4.1). We choose everything on validation and read the test split with the predictions written in advance.

On the test split, the graph branch recovers the source with an AUC of 0.903, against 0.611 for LayoutGKN retrained on the same task. Fusing it with the frozen image encoder adds $+0.013$, a gain that a second graph model matches; fusing it with a damage-trained vision head adds $+0.031$ (Fig. 1), and over four replicas the damage-trained heads exceed both controls by $+0.012$ to $+0.022$. The fusion gain follows an inverted U in the strength of the graph: with the frozen encoder it peaks at $+0.198$ when the two branches are equally strong, where even the frozen encoder passes both controls.

We make the following contributions:

- **Graph branch for incomplete plans.** Training on damaged–complete pairs and selecting on robustness, after the same search over GCN, GraphSAGE and GATv2, brings self-recovery under room removal to $0.904 \pm 0.001$ on test, 0.29 above a retrained LayoutGKN.
- **Ensemble controls and complementarity gain.** With a strong graph, a frozen image encoder adds no information beyond an ensemble effect, while damage-trained vision heads do; even then the stronger control, a second graph model fused with ours, reaches about half of the fusion gain, and the CG appears only where the graph has little left, at heavy damage and on small plans.
- **Relative strength.** Across 37 graphs, the fusion gain peaks when the branches are equally strong and shrinks near the ceiling, which reconciles a $+0.17$ gain in an earlier version of our pipeline with $+0.013$ here.
- **Protocol and negative results.** A self-recovery protocol with three damages, paired intervals, replicas and preregistered test readings, and the failures that shaped it: a rendering artifact, a head that learns only the damage it sees, and a graph that collapses with symmetric pairs.

## 2. Related Work

**Floor-plan similarity and retrieval.** Floor-plan retrieval needs a similarity that reflects structure beyond pixels. LayoutGMN [13] learns it with an attention-based graph matching network supervised by pixel IoU; its embeddings depend on the pair, so ranking a large gallery requires one forward pass per candidate. SSIG [21] combines IoU with a graph edit distance into a metric that needs no learning. LayoutGKN [20] moves the cross-graph interaction to a graph kernel on the final room embeddings, trained on RPLAN with a target based on graph edit distance. Unit Region Encoding [24] learns a compact geometry-aware representation from rasterised plans. These methods compare complete plans and evaluate on triplets or within the top results of the model; we rank the whole gallery from incomplete queries, and we retrain LayoutGKN on our task as a baseline.

**Incomplete queries.** On-the-fly sketch retrieval [3] retrieves a photo from a sketch that grows stroke by stroke and scores the area under the reciprocal-rank curve over completeness levels; our robustness AUC follows the same scheme, with a query that loses rooms or pixels and stays in the gallery modality. For floor plans, partial inputs appear in generation: Graph2Plan [9] retrieves RPLAN layout graphs that match a boundary and user constraints with hand-written rules, and SketchPlan within PlanCraft [23] replays the drawing process of RPLAN plans to produce partial sketches at several completeness levels.

**Multimodal floor-plan representations.** CrossOver [16] aligns RGB images, point clouds, CAD models, floor plans and text of 3D scenes in one embedding space and tolerates missing modalities. MMFE [1] maps floor plans of different modalities into a shared dense grid with a frozen DINOv3 backbone and a trained head. Both learn the fusion end to end and report the fused accuracy. Our fusion is a weighted sum of two cosine similarities, and our question is how much of its gain comes from information that only the image holds; ensemble controls answer it without changing the fusion.

**Encoders.** We propose no new encoder: both branches select among published ones. Frozen general-purpose encoders are the natural starting point on a new image domain, and we compare self-supervised encoders (DINOv2 [12], DINOv3 [17]), the predictive I-JEPA [2], vision–language encoders (SigLIP 2 [18], TIPSv2 [6]), the agglomerative C-RADIO [15] and the Perception Encoder [4] in its contrastive and spatially aligned variants, with generalised-

mean pooling [14], PCA whitening [10] and exact search with FAISS [7]. On the graph side we compare GCN [11], GraphSAGE [8] and GATv2 [5], trained with InfoNCE [19] on pairs that match a partial plan with its complete version.

## 3. Method

Let $\mathcal{G}$ be a gallery of floor plans, each available as a rendered image and as a structural record (room types, room boxes, typed adjacencies). A query $\tilde{q}$ is a plan $q \in \mathcal{G}$ with part of its content removed (Sec. 4.1). The task asks for a ranking of the whole gallery that places the source plan $q$ as high as possible. A vision encoder maps a plan to a unit vector $v \in \mathbb{R}^{768}$ and a graph encoder maps it to a unit vector $g \in \mathbb{R}^{128}$. Both branches index the same gallery and receive the same damaged queries, so we compare branches and fusions query by query (Fig. 2).

### 3.1. Late fusion and ensemble controls

**Fusion.** The fused descriptor concatenates the two L2-normalised vectors with a single weight $\alpha$,

$$z = \left[\sqrt{\alpha}\, v\,;\, \sqrt{1-\alpha}\, g\right], \\ \langle z_q, z_p \rangle = \alpha \langle v_q, v_p \rangle + (1-\alpha) \langle g_q, g_p \rangle, \tag{1}$$

so $\|z\| = 1$ and one exact inner-product search over 896-d vectors ranks the gallery by a convex combination of the two cosine similarities. At $\alpha = 1$ and $\alpha = 0$ the fusion reproduces the two branches, which we use as a correctness check. For a damaged query, both branches recompute the query vector from the *same* removed rooms; gallery vectors always describe complete plans. We choose $\alpha$ on the validation split over $\{0, 0.1, \ldots, 1\}$ with a rule fixed before the runs. The criterion is the robustness to room removal (Sec. 4.1), the only damage both branches can receive. Values whose paired difference from the best has a 95% bootstrap interval containing zero count as tied, and we take the central tied value. The rule gives $\alpha^* = 0.4$ in all four replicas, and the test split uses the weight fixed on validation. The fusion is static: one global weight, no per-query gate.

**Ensemble controls.** Fusing any two models with partly independent errors raises retrieval accuracy, so a fusion gain alone does not show that the image carries information the graph lacks. We therefore compare every vision–graph fusion with two *ensemble controls* that fuse the graph with a second model holding the *same* information: a second training of the same graph configuration (different initialisation and batches), and a graph encoder of another family trained with the same recipe. Each control selects its own weight on validation with the rule above. For a fusion $F$ of branches $A$ and $B$, the per-query gain is $\mathrm{gain}_F(q) = \mathrm{AUC}_F(q) - \mathrm{AUC}_b(q)$, where $b$ is the branch with the higher mean AUC.

**Complementarity gain.** We call *complementarity gain* the part of a vision–graph gain that no second graph model provides,

$$\mathrm{CG} = \overline{\mathrm{gain}}_{V+G} - \overline{\mathrm{gain}}_{C^\star}, \tag{2}$$

where $C^\star$ is the ensemble control with the larger mean gain, a conservative choice. We compute CG on paired queries with bootstrap intervals. $\mathrm{CG} > 0$ means the vision branch contributes information beyond an ensemble effect; the remaining part $\overline{\mathrm{gain}}_{C^\star}$ is what a second graph model would already deliver. The two parts come from different fusions, so they summarise the gain and do not split one fusion additively.

### 3.2. Graph branch

**Room graph.** We build each graph from the RPLAN record [22], not from the image. Each room becomes a node whose 19 features concatenate a one-hot encoding of the 13 room types with the centre, width, height, area and aspect ratio of the room box on the $256 \times 256$ RPLAN grid; we clip the aspect ratio at its 99th percentile and standardise the geometric features. Each adjacency becomes an undirected edge with a one-hot encoding of the 10 RPLAN relation types.

**Encoder.** The encoder $W$ stacks three GATv2 layers [5] with hidden size 256, one attention head and edge features in the attention. We sum-pool the node states and concatenate the sum-pooled input features; on the one-hot part this skip connection equals the room-type histogram. A linear layer projects the result to 128 dimensions, followed by L2 normalisation (317,952 parameters).

**Training for incompleteness.** We train with the symmetric InfoNCE loss [19], in-batch negatives, temperature $\tau = 0.1$, batch size 256, learning rate $10^{-3}$, weight decay $10^{-5}$ and no dropout, on the training split only. Positive pairs are *asymmetric*: one view is the complete plan, the other removes a fraction $f \sim \mathcal{U}[0.25, 0.75]$ of its rooms together with their edges. The pair thus asks the encoder to map a partial plan onto its complete version, the retrieval task itself. We select the checkpoint on retrieval, not on the loss: after each epoch we measure the robustness to room removal on 2,000 validation queries disjoint from the evaluation queries, with the whole gallery, over 600 epochs. The selected epoch falls between 550 and 592 in all four replicas, so the reported robustness of the graph is a lower bound.

**Configuration search.** We chose $W$ with the same recipe for GCN [11], GraphSAGE [8] and GATv2. Starting from a reference configuration per encoder, we changed one factor

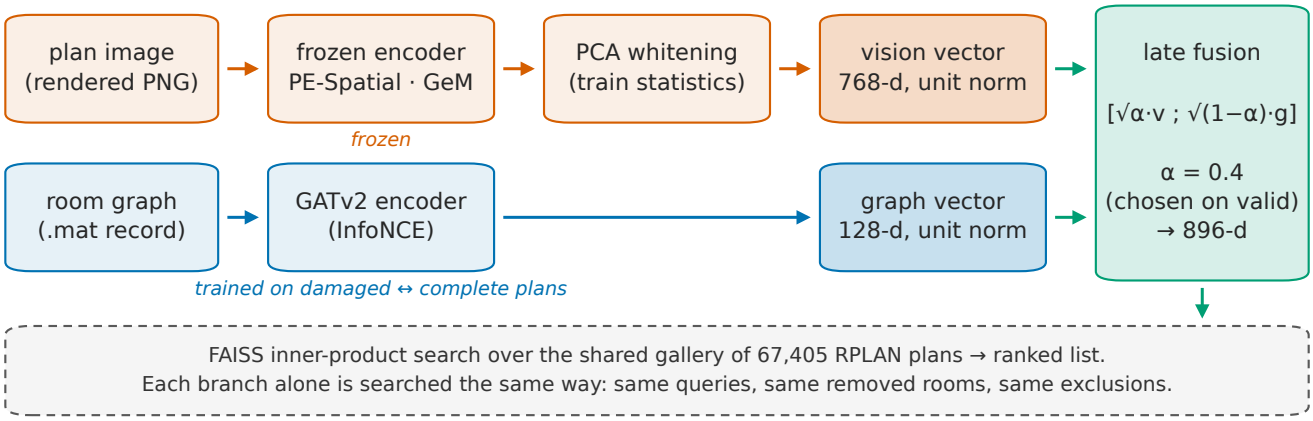



Figure 2. **FusePlan pipeline.** The vision branch embeds the rendered plan with a frozen encoder and whitening; the graph branch embeds the room graph with a GNN trained to map partial plans onto complete ones. Late fusion concatenates the two unit vectors with weights $\sqrt{\alpha}$ and $\sqrt{1-\alpha}$ ($\alpha = 0.4$, chosen on validation), so one inner product scores both. The damage-trained vision variants (Sec. 3.3) add a trained head to the vision branch and leave the rest unchanged. An exact inner-product FAISS search ranks the gallery of 67,405 plans with the fused vector; each branch alone is searched the same way, with the same queries and removed rooms.

at a time: temperature, skip connection, rotation and reflection augmentation, a lost-neighbour marker, depth, width, pooling, aggregator and attention heads. We then combined the changes that improved robustness in all four seeds. We train each of the 37 candidates with 4 seeds and score it on validation only (Tab. 3). $W$ is the combination for GATv2: $\tau = 0.1$, no rotations or reflections, three layers, width 256 and one head.

### 3.3. Vision branch

**Frozen encoder.** The vision branch uses the spatially aligned Perception Encoder [4] (PE-Spatial, ViT-B/16) without training. We resize the rendered plan so that its longer side has 224 pixels, keeping the aspect ratio, and pad it to a square with the white background. The descriptor is a generalised mean with $p = 3$ over the last-layer patch tokens, computed on their positive part: the implementation clamps the post-LayerNorm tokens at $10^{-6}$ before pooling, which differs from standard GeM [14].

**Whitening and indexing.** We apply PCA whitening [10]: mean and covariance come from training-split plans only, and every descriptor $x$ maps to $\Lambda^{-1/2}U^{\top}(x-\mu)$, keeping all 768 components, followed by L2 normalisation. An exact inner-product FAISS index [7] stores the gallery. Whitening matters because the frozen descriptors of the synthetic renderings collapse into a narrow cone; it improves robustness in all 26 encoder–pooling pairs we tested (Tab. 4). PE-Spatial with this pooling and whitening is the most robust of 52 frozen configurations over 8 encoders [2, 4, 6, 12, 15, 17, 18], selected on validation.

**Damage-trained heads.** To test whether a vision branch trained on the damage carries more information, we add two heads on top of the frozen encoder, trained with InfoNCE ($\tau = 0.07$, batch 4,096) on training plans, with self-supervised positives and no relevance labels. The *single-damage head* is an MLP ($1024 \to 256$, GELU) applied to gallery and queries; it learns to match a plan with rooms removed to the complete plan, and we re-estimate the whitening on its outputs. The *multi-damage head* acts on queries only: it adds a residual MLP ($768 \to 1024 \to 768$, last layer initialised at zero) to the whitened frozen vector, so the gallery stays frozen; it trains on room removal, crop and patch in turn. Both heads select their epoch on 1,000 validation queries disjoint from the evaluation queries.

## 4. Experiments and Results

### 4.1. Metrics

**Incomplete queries.** A query is a validation or test plan from which we remove a fraction $f \in \{0.25, 0.5, 0.75\}$ of the content with one of three damages, while the gallery keeps the complete plans. *Room removal* deletes $\mathrm{round}(f\,n)$ of the $n$ rooms, drawn with a per-query seed shared by the two branches, so both remove the same rooms. In the image we fill the removed rooms with the background colour and erase their walls, so the query keeps only the outline of the hole; in the graph we keep the subgraph induced by the remaining rooms. Rooms differ in size, so the removed area is on average 0.29, 0.52 and 0.71 of the plan at the three levels. *Crop* blanks one axis-aligned rectangle covering a fraction $f \pm 0.05$ of the plan pixels, and *patch* blanks whole ViT patches until they cover a fraction $f$ of the plan. Crop and patch act on pixels, so only the vision

branch receives them.

**Robustness metric.** We measure *self-recovery*: the gallery still contains the intact source plan, and a query scores the reciprocal rank $\mathrm{rr}(q, f)$ of its source within the top 100, and 0 beyond. The robustness of a system on a damage is the area under the self-recovery curve, $\mathrm{AUC}(q) = \frac{1}{3}\sum_f \mathrm{rr}(q, f)$, averaged over queries, as in the m@B metric of on-the-fly sketch retrieval [3]. Branches and fusions compete on room removal, the damage they share. To select the vision configuration we average the AUC of the three damages, so that no configuration wins by exploiting one damage. We exclude $f = 0$: about 3% of the RPLAN images have a pixel-identical twin, which caps the undamaged self-recovery near 0.97 for every system. Self-recovery needs no relevance label, so the circularity discussed below does not affect it.

**Statistics.** We pair every comparison over the same queries, gallery and removed rooms, with 95% percentile bootstrap intervals over queries ($B = 10{,}000$). Across four replicas, which change the graph training and the removed rooms, we call an effect *above the training noise* when it has the same sign in 4/4 replicas and its mean exceeds twice the standard deviation across replicas. For equivalence we use a margin of $\pm 0.005$, fixed after the first replica and before the others.

**Per-axis relevance (descriptive).** To describe what a system retrieves on complete plans, we score its ranking with nDCG@10 on three relevance axes computed from the metadata, each in $[0, 1]$. *Composition* is the weighted Jaccard similarity of the 13-bin room-type histograms; *topology* is the weighted Jaccard similarity of the 91-bin histograms of adjacent type pairs; *geometry* averages the similarities of footprint area, footprint aspect ratio and per-type area distribution. A random ranking already scores 0.740, 0.466 and 0.870 on the three axes (test), so we also report the gain normalised to this floor, $(s - s_0)/(1 - s_0)$. These scores are circular for both branches and we never use them to select. The graph receives the room types and adjacencies that composition and topology measure, and the footprint is the union of the room boxes in every record we checked. The renderings encode the room types as colours, with 13 types collapsed into 6 colours. Per-axis differences between the branches therefore compare a fine and a coarse reading of the same information.

### 4.2. Dataset and experimental details

**Dataset.** RPLAN [22] provides, for each residential plan, a colour rendering ($1167 \times 875$ pixels, rooms coloured by category) and a record with 13 room types, room boxes and typed adjacencies. The gallery contains the 67,405 plans that have both an image and a record (48 images lack a record) and both branches index it in the same order. Following the official split, the gallery holds 47,126 training, 10,152 validation and 10,127 test plans. Training and whitening statistics use training plans only, while the gallery always contains every plan, as in a search engine. Graphs are small: 4 to 8 rooms, 6.8 on average.

| | test, seed 42 | | | | test, 4 rep. |
|---|---|---|---|---|---|
| System | $f$=.25 | .50 | .75 | AUC | AUC |
| hist | .003 | .001 | .000 | .001 | – |
| Frozen vision | .822 | .304 | .064 | .396 | .392±.005 |
| Single-damage head | .896 | .530 | .173 | .533 | .531±.001 |
| Multi-damage head† | – | – | – | .653 | – |
| LayoutGKN† | .940 | .676 | .217 | .611 | – |
| Graph $W$ | .963 | .954 | .793 | .903 | .904±.001 |
| $W$ + frozen vision | .961 | .955 | .832 | .916 | .916±.001 |
| $W$ + single-damage head | .963 | .958 | **.881** | .934 | .932±.001 |
| $W$ + multi-damage head | – | – | – | **.941** | **.940**±.001 |

Table 1. **Room removal on the test split.** Self-recovery per level and its AUC; the last column is the mean ± std over four replicas (graph training and removed rooms). $W$ beats the retrained LayoutGKN by 0.29 AUC, and every fusion improves on $W$, almost entirely at $f = 0.75$. †One training.

**Protocol.** We make every choice (encoder configuration, checkpoint, fusion weight, control weights) on 2,000 validation queries drawn with a fixed seed, and evaluate on 2,000 test queries drawn in the same way. Before each test reading we wrote the hypotheses, the predicted outcomes and the decision rules, then read the test split once with all weights fixed. Earlier stages of the project also read the test split, with a weaker graph branch; those readings select nothing reported here, and we disclose them in Sec. 5. Replicas use seeds 42, 100042, 200042 and 300042.

**Baselines.** *hist* ranks plans by the L2 distance between room-type histograms, without training. *LayoutGKN* [20] is the closest published graph method; we retrain its GKN-asym network with its upstream hyperparameters on our pairs (damaged plan, complete plan) over the whole training split for 300 epochs, with edges from the record adjacencies, and score it with our protocol. The same network with random weights reaches an AUC of 0.0001, so the retrained model learns the task.

### 4.3. Results

**A graph trained on incomplete plans comes close to the ceiling.** Table 1 and Fig. 3 report the test results. $W$ reaches an AUC of 0.903 and still ranks the source near the top with three quarters of the rooms removed (0.793).

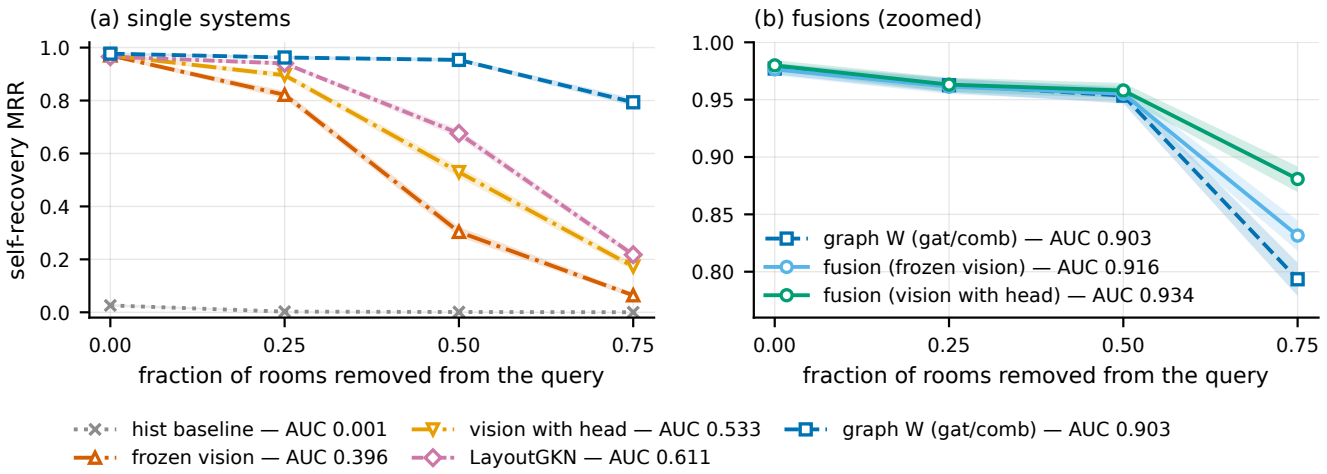



Figure 3. **Self-recovery under room removal on the test split (seed 42), with 95% bootstrap bands.** (a) $W$ keeps the source near the top up to half of the rooms removed, where LayoutGKN and the vision branches drop. (b) The fusions match $W$ up to $f = 0.5$ and gain at $f = 0.75$, most with the single-damage head.

Retrained on the same pairs, LayoutGKN scores 0.611, a paired difference of $-0.292$ $[-0.302, -0.282]$. LayoutGKN stays close to $W$ when a quarter of the rooms is missing (0.940 vs 0.963) and drops when most are missing (0.217 vs 0.793). The training-free histogram finds almost nothing (0.001), and the frozen vision branch reaches 0.396. The standard deviation of $W$ across four replicas is 0.001 on the test split, so the graph result does not depend on one training.

**Fusing with the frozen vision branch adds $+0.013$.** The fusion with the frozen vision branch improves on $W$ by $+0.013$ $[+0.009, +0.017]$ on the test split, with a positive interval in all four replicas ($+0.0125 \pm 0.0012$) and a test value within 0.006 of validation in each. Almost all of the gain comes at $f = 0.75$ (0.832 vs 0.793), the only level where $W$ leaves room. With the single-damage head the gain more than doubles, to $+0.031$ $[+0.026, +0.035]$, and with the multi-damage head the fused AUC reaches $0.940 \pm 0.001$. The fusions also exceed a per-query oracle that picks the better branch for each query (0.914 for the frozen fusion), because summing similarities can rank the source above both branches on the same query.

**With a strong graph, only a damage-trained vision branch adds information.** Table 2 and Fig. 4 compare each fusion gain with the ensemble controls. Over four replicas, fusing $W$ with a second training of itself gains $+0.0163$ and fusing it with the SAGE graph gains $+0.0145$, both more than the frozen vision fusion ($+0.0125$). Over four replicas the frozen vision gain is equivalent to the SAGE control ($-0.0020$ $[-0.0045, +0.0005]$, within $\pm 0.005$) and slightly below the second copy of $W$ ($-0.0038$ $[-0.0067, -0.0010]$, negative in 4/4 replicas). The single-damage head exceeds both controls by $+0.012$ $[+0.008, +0.015]$ and $+0.013$ $[+0.011, +0.016]$, and the multi-damage head by $+0.020$ $[+0.017, +0.023]$ and $+0.022$ $[+0.019, +0.025]$, all positive in 4/4 replicas and above the training noise. The single-damage result was a pre-registered confirmatory test on seed 42 ($+0.014$ $[+0.008, +0.020]$ and $+0.015$ $[+0.011, +0.020]$) before the replicas. Even when the complementarity gain is positive, the control accounts for 32–58% of the fusion gain, so reporting the raw gain overstates the contribution of the image by about a factor of two.

| Fusion | gain | gain $C^\star$ | CG [95% CI] | CG share |
|---|---|---|---|---|
| *Strong graph $W$ (AUC 0.90), mean of 4 replicas* | | | | |
| + frozen vision | .0125 | .0163 | −.0038 [−.0067, −.0010] | – |
| + single-damage | .0279 | .0163 | +.0116 [+.0085, +.0148] | 42% |
| + multi-damage | .0366 | .0163 | +.0203 [+.0171, +.0235] | 55% |
| *Graphs as strong as the vision branch, seed 42* | | | | |
| gat/t05 + frozen | .1978 | .0912 | +.1066 [+.0937, +.1195] | 54% |
| gat/ref + frozen | .1516 | .0720 | +.0797 [+.0684, +.0911] | 53% |
| gat/t05 + single | .1775 | .0912 | +.0863 [+.0761, +.0963] | 49% |
| gat/ref + single | .2229 | .0720 | +.1509 [+.1406, +.1613] | 68% |

Table 2. **Complementarity gain on the test split.** gain: fused AUC minus the better branch; gain $C^\star$: the same for the stronger ensemble control (second copy of $W$ in the top block, the SAGE graph with the same change in the bottom block); CG = gain − gain $C^\star$ (Eq. (2)). The frozen vision branch adds nothing beyond an ensemble effect with $W$, while damage-trained heads and weaker graphs leave a positive CG.

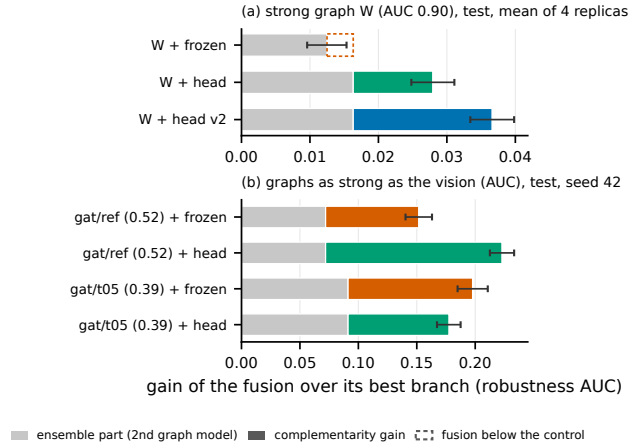



Figure 4. **About half of each fusion gain is an ensemble effect (test).** Grey: gain of the stronger ensemble control; colour: complementarity gain with 95% interval. With $W$ the frozen vision fusion falls below the control (dashed).

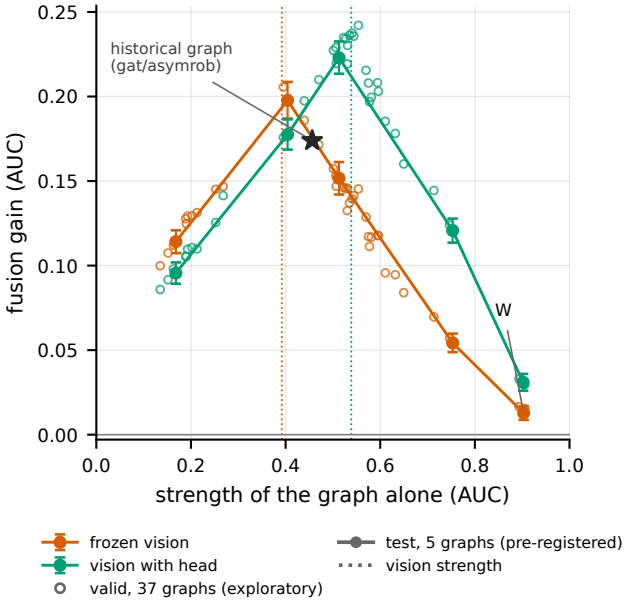



Figure 5. **The fusion gain peaks when the two branches are equally strong.** Gain over the better branch against the AUC of the graph alone: 37 graphs on validation (open circles, exploratory) and 5 on test (filled, pre-registered, 95% intervals); dotted lines mark the strength of each vision branch. The historical graph of an earlier pipeline (star) and $W$ lie on the same curve.

**The gain depends on the relative strength of the branches.** Figure 5 fuses the 37 graphs of the configuration search, from the weakest to $W$, with the two vision branches. The gain follows an inverted U. With graphs weaker than the vision branch it grows with the graph; it peaks when the branches are equally strong ($+0.198$ on test for a graph at 0.404 with the frozen vision branch); it shrinks as the graph approaches the ceiling ($+0.054$ at 0.753, $+0.013$ for $W$). An earlier version of our pipeline used a graph at 0.456 on validation and measured a fusion gain of $+0.174$; that point lies on the same curve, so the large historical gain and the small gain with $W$ do not contradict each other. The single-damage head lowers the gain with the weakest graph ($-0.019$ against the frozen branch), where it raises the reference branch, and raises it with medium and strong graphs ($+0.071$, $+0.066$, $+0.018$). All nine pre-registered comparisons on the test split go in the predicted direction, in each of four seeds. With graphs as strong as the vision branch, even the frozen vision branch passes both controls by $+0.08$ to $+0.16$ (Tab. 2, bottom).

**The image helps where the graph has little left.** Figure 6 locates the complementarity gain of the fusions with $W$ on the test split. With a quarter or half of the rooms removed the CG is zero for all three vision branches, with intervals that contain zero; at $f = 0.75$ it reaches $+0.046$ $[+0.031, +0.061]$ for the single-damage head and $+0.071$ $[+0.054, +0.087]$ for the multi-damage head. On the 639 of 2,000 queries where $W$ misses the first rank at some level, the heads gain $+0.052$ and $+0.073$ over the SAGE control, while the frozen branch gains $+0.004$ $[-0.007, +0.016]$. On the 154 plans with at most five rooms the CG grows to $+0.067$ and $+0.103$, and the frozen branch also exceeds zero ($+0.028$ $[+0.002, +0.053]$). We expected a larger CG for plans that share their room graph with many others; the data refute it. These groups overlap, since small plans and heavy damage both leave the graph with few rooms, and we apply no correction for the 39 group comparisons, so we read them as a description of one seed.

**Takeaway.** The vision branch contributes information of its own when the graph has little left: at heavy damage, on small plans, and with graphs no stronger than the image. With a graph near the ceiling, a frozen image encoder adds no more than a second graph model, and the fusion gain alone would have suggested otherwise.

#### 4.3.1. Ablation

**Graph branch.** Table 3 shows that a temperature of 0.1 is the strongest single change for all three encoders ($+0.10$, $+0.17$ and $+0.23$ AUC over the reference), while $\tau = 0.5$ and removing the skip from the input features hurt everywhere. Combining the useful changes matters more than any single one, above all at heavy damage: at $f = 0.75$ the GATv2 combination recovers 0.808 against 0.378 for $\tau = 0.1$ alone. $W$ beats the SAGE combination by $+0.0094$ $[+0.0064, +0.0125]$ on validation and by $+0.011$ $[+0.008, +0.014]$ on test over four replicas; GCN stays far behind. The most robust graphs are not the best on complete

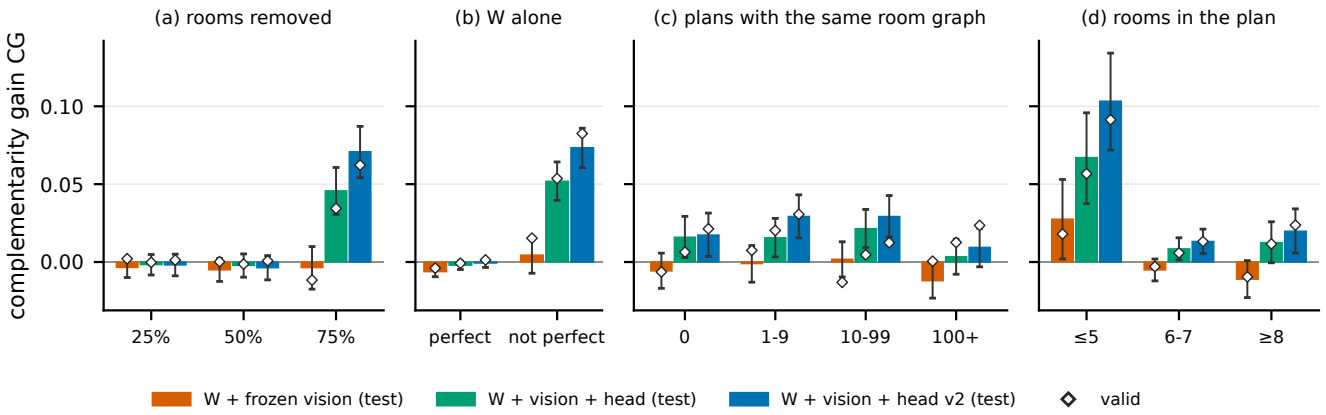



Figure 6. **The complementarity gain appears where the graph keeps little information.** CG of the three fusions with $W$ by query group (bars: test, seed 42; diamonds: validation; 95% intervals). (a) Fraction of rooms removed. (b) Whether $W$ alone ranks the source first at every level. (c) Number of gallery plans with the same typed room graph. (d) Rooms in the plan. Post-hoc analysis: groups and hypotheses written before computing them, on splits already read.

| Change from the reference | GCN | SAGE | GATv2 |
|---|---|---|---|
| reference ($\tau = 0.3$) | .163±.003 | .543±.007 | .516±.022 |
| $\tau = 0.1$ | .259±.006 | .717±.004 | .745±.010 |
| $\tau = 0.5$ | .133±.005 | .442±.011 | .384±.008 |
| no rotation/reflection | .274±.005 | .634±.007 | .612±.005 |
| 3 layers | .208±.008 | .651±.005 | .587±.009 |
| hidden 256 | .195±.005 | .583±.009 | .577±.004 |
| no skip from input | .159±.007 | .522±.012 | .510±.011 |
| combination | .564±.007 | .897±.003 | **.907**±.004 |

Table 3. **Graph configuration search (validation, room-removal AUC, mean ± std over 4 seeds).** Each row changes one factor of the reference; the last row combines, per encoder, the changes that improved robustness in 4/4 seeds. The GATv2 combination is $W$.

plans: the GCN combination reaches a topology nDCG@10 of 0.778 against 0.647 for $W$.

*(a) Frozen encoders, validation, mean AUC of three damages*

| Encoder | pooling | raw | whitened |
|---|---|---|---|
| PE-Spatial [4] | GeM | .293 | **.525** |
| C-RADIOv2 [15] | summary | .129 | .516 |
| I-JEPA [2] | mean | .107 | .426 |
| DINOv3 [17] | mean | .181 | .386 |
| DINOv2 [12] | [CLS] | .173 | .381 |
| TIPSv2 [6] | [CLS] | .111 | .317 |
| SigLIP2 [18] | mean | .079 | .292 |
| PE-Core [4] | mean | .015 | .281 |

*(b) Vision variants, test: AUC per damage*

| | rooms | crop | patch |
|---|---|---|---|
| Frozen (PE-Spatial) | .397 | .654 | .512 |
| Single-damage head | .532 | .515 | .412 |
| Multi-damage head | **.653** | **.820** | **.672** |

Table 4. **Vision branch.** (a) Best configuration per encoder, with and without whitening. (b) The single-damage head gains on the damage it trains on and loses on the other two; the multi-damage head gains on all three.

**Vision branch.** Whitening improves robustness in all 26 encoder–pooling pairs (Tab. 4a): without it the descriptors of the synthetic renderings collapse and damaged queries match almost any plan. The choice of encoder matters more than pooling or resolution, and newer or larger encoders do not win: TIPSv2, the best encoder on complete plans in our benchmark, ranks sixth under damage, and the ViT-H I-JEPA trails two ViT-B models. The best configuration on each single damage changes from damage to damage, which is why we select on their average. The multi-damage head reaches a mean AUC of 0.715 on the three test damages against 0.521 for the frozen branch, and improves each damage with a positive interval (Tab. 4b).

**Negative results.** We report the outcomes that contradicted our expectations, because each one changed the protocol. *(i) A head learns the damage it sees.* The single-damage head improves room removal by $+0.136$ on test and loses $-0.139$ on crop and $-0.101$ on patch, a pre-registered transfer hypothesis that we refuted on two encoders. *(ii) A rendering artifact inflated the vision robustness.* Our first room-removal damage erased the dark perimeter walls and kept the lighter interior walls, so each removed room remained an empty outlined cell. On the same queries and rooms, the artifact raised the AUC of a frozen DINOv3 configuration from 0.304 to 0.536 ($+0.336$ at $f = 0.5$),

and a head trained on that rendering exploited it. We found the artifact because we had pre-registered a check on other damages; the three-damage criterion comes from this lesson. *(iii) The graph needs the right pairs and the right checkpoint.* Trained on symmetric augmented pairs, every GNN ranks the source of a damaged query out of reach (AUC below 0.03) while recovering complete plans; asymmetric pairs fix this. Selecting the checkpoint on complete plans picks early, erratic epochs: on one training trajectory it gives 0.107, against 0.457 when selecting on robustness. For the graph, the InfoNCE validation loss even ranks configurations in the opposite order of retrieval, because in-batch negatives often share the composition of the anchor. *(iv) Refuted predictions.* We predicted a SAGE configuration to win the search, the frozen fusion to gain more than $+0.12$ with $W$, and the frozen vision branch to be equivalent to the second copy of $W$; all three failed. In the second round, 43 of 45 pre-registered predictions held, and the two failures concern this last point.

#### 4.3.2. Qualitative results

Figure 1 shows a validation query with half of its rooms removed. $W$ ranks first a near-identical plan and the source second; the single-damage head misses the source in its top 100; the fusion ranks the source first. Its top-5 lists show the mechanism we observe across queries: one branch keeps the source near the top, and the other demotes the competitors that outrank it. On complete plans the fusions change little. On test, the frozen fusion scores 0.825, 0.662 and 0.953 nDCG@10 on composition, topology and geometry, against 0.818, 0.651 and 0.949 for $W$. The fusion thus covers 33%, 37% and 64% of the room above the random floor. These per-axis numbers are circular and descriptive, and they show that complete-plan retrieval is far from solved even when self-recovery is near the ceiling.

## 5. Conclusions

A graph encoder trained on damaged–complete pairs and selected on robustness retrieves a plan from half of its rooms almost as well as from the complete plan. Fusing it with an image encoder adds information of its own under two conditions: the image branch trains on the damage, or the graph is no stronger than the image. Ensemble controls make this visible, and the complementarity gain gives it a number: even when positive, it is about half of the fusion gain. We recommend reporting fusion gains next to ensemble controls whenever two branches see related information.

**Limitations.** We use RPLAN only, with synthetic damages: a removed room leaves the outline of its hole in the image, and crop and patch reach only the vision branch, so we compare branches and fusions on room removal alone. The graph comes from exact records, not from parsed images, and the renderings encode room types as colours, so both branches read the same information at different granularity and every per-axis relevance score is circular. Training budgets bind: $W$ selects epochs 550–592 of 600 and LayoutGKN, trained once with upstream hyperparameters, stops at 300, so both robustness values are lower bounds. The fusion uses one global weight; a per-query gate remains untested, although a hard gate cannot exceed the per-query oracle, which the static fusion already exceeds. We trained the multi-damage head once and judged it only on the damages it trained on. Fusion gains with $W$ are small in absolute terms, although paired and stable across replicas. The analysis of where the complementarity gain appears is post hoc and uses one seed. Finally, earlier stages of this project read the test split: an exploratory benchmark of vision and graph configurations, three vision variants on complete plans, and two pre-registered readings of a pipeline with a weaker graph. All numbers in this paper come from two later pre-registered readings, with every weight fixed on validation.

**Future work.** Graphs parsed from images would remove the circularity; external datasets would test generality without retraining; a learned or per-query fusion, a fourth damage outside the training set of the heads, and human relevance judgements are the next steps.

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